CMPS 261 Progress Report: NASA Kepler Exoplanet Candidate Classification

Group 4: Joe Habre and Joy Habib Najem — **Track A:** 4.1 Tabular (NASA KOI)
GitHub Repository: `https://github.com/Joeehabre/CMPS-261.git`

1. Problem Statement

The NASA Kepler Objects of Interest (KOI) cumulative dataset contains $N = 9,564$ transit events observed by the Kepler Space Telescope across 50 tabular attributes describing orbital parameters, transit signal geometry, and host-star properties. Each observation is assigned an archive disposition (`koi_disposition`): `CONFIRMED` (2,293), `CANDIDATE` (2,248), or `FALSE POSITIVE` (5,023). Our objective is to build and evaluate machine-learning models that distinguish promising exoplanet observations from astrophysical and instrumental false positives (such as eclipsing binary stars or background stellar contamination). We formulate the primary task as binary classification: **Potential Exoplanet** (`CONFIRMED` ∪ `CANDIDATE`, $y = 1$, 4,541 instances, 47.48%) versus **False Positive** (`FALSE POSITIVE`, $y = 0$, 5,023 instances, 52.52%). Because unconfirmed candidates may contain residual false positives, we also define a secondary strict benchmark restricted to verified labels (`CONFIRMED` vs. `FALSE POSITIVE`, $N = 7,316$) to test label-noise sensitivity.

Changes Since Initial Proposal: While our core problem formulation remains unchanged from the initial proposal, we introduced two methodological refinements during baseline development: (1) a formal pre-vetting feature audit that eliminates all post-classification diagnostic flags (`koi_fpflag_*`) and pipeline disposition scores, and (2) **host-star grouped partitioning** (`GroupShuffleSplit` and `GroupKFold` on star ID `kepid`) so multi-planet stellar systems never straddle training and validation splits.

2. Data Preprocessing

1. **Target Leakage Audit:** Several columns in the cumulative KOI table are populated only *after* scientists or automated vetting pipelines classify an object. Including them inflates accuracy above 98% via target leakage. We audited all 50 attributes and removed 13 columns prior to modeling:
   - *Post-Vetting Leakage Columns (6):* `koi_pdisposition` (preliminary disposition), `koi_score` (vetting confidence score in $[0, 1]$), and the four diagnostic false-positive flags: `koi_fpflag_nt` (not transit-like), `koi_fpflag_ss` (stellar eclipse), `koi_fpflag_co` (centroid offset), and `koi_fpflag_ec` (ephemeris match contamination).
   - *Identifiers & Non-Predictive Metadata (5):* `rowid`, `kepid` (retained solely as a grouping key), `kepoi_name`, `kepler_name` (assigned only to confirmed planets, leaking $y = 1$), and `koi_tce_delivname`.
   - *Unobserved Columns (2):* `koi_teq_err1` and `koi_teq_err2` (100% missing across all 9,564 rows).
2. **Physical Feature Space & Engineering:** The resulting clean input matrix X comprises 36 raw pre-vetting physical features across transit parameters (`koi_period`, `koi_duration`, `koi_depth`, `koi_impact`, `koi_model_snr`, `koi_time0bk`), planetary estimates (`koi_prad`, `koi_teq`, `koi_insol`, `koi_tce_plnt_num`), stellar host properties (`koi_steff`, `koi_slogg`, `koi_srad`), their upper/lower measurement uncertainties (`*_err1`, `*_err2`), and sky coordinates (`ra`, `dec`, `koi_kepmag`). We engineered a physics-informed transit consistency ratio comparing observed transit depth δ (`koi_depth`) against the geometric planet-to-star area ratio $(R_p/R_*)^2$, as grazing eclipsing binaries violate $\delta \approx (R_p/R_*)^2$. Heavily right-skewed positive features (`koi_period`, `koi_depth`, `koi_prad`, `koi_insol`, `koi_model_snr`, `koi_duration`) undergo $\log(1 + x)$ transformation.
3. **Grouped Splitting, Imputation, and Scaling:** Because 9,564 KOIs orbit 8,214 unique stars (`kepid`), random row splitting leaks host-star parameters (`koi_steff`, `koi_srad`) between train and test sets. We partitioned the data into an 80% development set (7,651 KOIs) and a 20% held-out test set (1,913 KOIs) grouped by `kepid`, with 5-fold `GroupKFold` cross-validation on the development set. For linear, Random Forest, and neural models, missing entries (3.8% overall, including 363 rows missing fitted stellar radii) are imputed within each fold using median imputation plus binary missingness indicators (`add_indicator=True`), followed by `StandardScaler`. For XGBoost, we also evaluate native missing-value routing, which learns the split direction that maximizes split gain during training.

3. Evaluation Metrics

- **ROC-AUC and PR-AUC (Ranking Performance):** Because follow-up radial-velocity spectroscopy is resource-constrained, astronomers prioritize candidates by predicted probability. ROC-AUC measures global ranking separability across all thresholds, while Precision-Recall AUC (PR-AUC) directly captures precision trade-offs at high recall.
- **F1-Score, Precision, and Recall (Classification Trade-off):** False Negatives discard real exoplanets, whereas False Positives waste telescope observation time on stellar binaries. Reporting Precision, Recall, and their harmonic mean (F1) at threshold $\tau = 0.5$ quantifies both error types.
- **Brier Score & Expected Calibration Error (Confidence Trustworthiness):** Because valid probabilities

in $[0, 1]$ are not automatically calibrated, we measure Brier Score and 10-bin Expected Calibration Error (ECE = $\sum_{b=1}^{10} \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$) to verify whether predicted confidence matches empirical accuracy.

4. Classical Baseline (Level 1)

We constructed a structured Level 1 baseline ladder evaluated via 5-fold `GroupKFold` cross-validation:

1. **Linear Baseline (Logistic Regression):** A linear model ($z = w^\top x + b$) with Sigmoid activation trained under Binary Cross-Entropy loss and L_2 regularization tuned over $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ ($C^* = 0.1$).
2. **Bagging Baseline (Random Forest):** An ensemble of $B = 500$ decision trees trained via bootstrap aggregation with random feature subsets $K = \sqrt{N}$ at each split (`max_depth=16`) to reduce variance.
3. **Boosted Baseline (Tuned XGBoost):** Sequential gradient-boosted decision trees trained on first-order gradients (g_i) and second-order Hessians (h_i) of the log-loss objective, optimizing split gain $\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$. Hyperparameters were tuned over learning rate $\eta \in \{0.01, 0.03, 0.05, 0.1\}$, `max_depth` $\in \{4, 6, 8\}$, L2 leaf regularization $\lambda \in \{1, 5, 10\}$, split penalty $\gamma \in \{0, 0.5, 1.0\}$, row subsampling (0.8), and column subsampling (0.8).

Model (5-Fold Grouped CV)	ROC-AUC	PR-AUC	F1-Score	Precision	Recall	ECE (↓)	Fi
Logistic Regression ($L_2, C = 0.1$)	0.854 ± 0.009	0.838 ± 0.011	0.781 ± 0.010	0.768	0.795	0.041	
Random Forest (500 trees, $d = 16$)	0.916 ± 0.007	0.908 ± 0.008	0.839 ± 0.008	0.831	0.847	0.034	
XGBoost (Tuned Baseline)	$\mathbf{0.928 \pm 0.006}$	$\mathbf{0.921 \pm 0.007}$	$\mathbf{0.854 \pm 0.007}$	**0.846**	**0.862**	**0.026**	
ResNet-MLP (Uncalibrated, $T = 1$)	0.911 ± 0.008	0.903 ± 0.009	0.834 ± 0.008	0.827	0.841	0.056	
ResNet-MLP (Temp. Scaled, $T^* = 1.42$)	0.911 ± 0.008	0.903 ± 0.009	0.834 ± 0.008	0.827	0.841	0.024	

5. Sophisticated Model (Level 2)

- **Architecture & Justification:** For Level 2, we implemented a deep tabular **Residual Multi-Layer Perceptron (ResNet-MLP)** with skip connections ($y = F(x) + x$), Batch Normalization, ReLU hidden activations, Dropout ($p = 0.25$), and Sigmoid output activation, trained with mini-batch AdamW ($\alpha = 10^{-3}$, batch size 128) and Early Stopping (patience 12). While decision trees partition continuous physical laws (such as Kepler's Third Law $P^2 \propto a^3/M_*$) using axis-aligned step functions, a residual neural network learns smooth, continuous hierarchical representations across coupled orbital and stellar features while skip connections preserve gradient flow. To address neural network overconfidence, we freeze trained weights and apply post-hoc **Temperature Scaling** ($\hat{p} = \sigma(z/T)$), optimizing scalar T on a held-out calibration split.
- **Current Progress & Preliminary Results:** Our ResNet-MLP substantially outperforms Logistic Regression (+0.057 ROC-AUC), confirming strong non-linear representation learning, and Temperature Scaling ($T^* = 1.42 > 1$) softens overconfident logits, cutting ECE from 0.056 to 0.024 without altering classification ranking or accuracy. However, the standalone neural network currently trails tuned XGBoost by 0.017 ROC-AUC while requiring 7.3× longer training time per fold. We are next implementing feature tokenization with self-attention (**FT-Transformer**) and evaluating a calibrated **XGBoost + ResNet-MLP ensemble** to test whether combining axis-aligned tree splits with smooth neural manifolds improves over either family alone.

6. Remaining Plan and Risks (Until November 20)

- **Remaining Experiment Schedule:**
  - *Nov 6–Nov 10 (Level 2 Tuning & Ensembling):* Tune FT-Transformer and ResNet-MLP hyperparameters (depth, weight decay, cosine annealing) and evaluate probability-calibrated ensembles on the held-out test set.
  - *Nov 11–Nov 14 (Split Sensitivity & Leakage Ablations):* Quantify sensitivity to evaluation design by comparing (i) Host-Star Grouped Split vs. Random Split (measuring host-star leakage inflation) and (ii) Binary Candidate grouping vs. Strict Confirmed-only disposition, alongside a feature ablation quantifying the exact accuracy jump when post-hoc flags (`koi_fpflag_*`) are included.
  - *Nov 15–Nov 20 (Failure Mitigations, Complexity Trade-offs & Final Deliverables):* Compute parameter counts, training/inference latency, and performance-vs.-complexity Pareto curves; finalize the poster, LaTeX report, and GitHub model weights.
- **Early Observations on Model Failures:** Stratified error analysis of our tuned XGBoost baseline (14.6% overall error) highlights four systematic failure regimes: (1) **Weak Signals (`koi_model_snr` < 12):** misclassification rises to 28.4% (vs. 8.1% for SNR > 50) as shallow transits merge with photometric noise; (2) **Grazing Eclipsing Binaries (`koi_impact` > 0.9, $R_p \in [8, 22]\, R_\oplus$):** V-shaped stellar eclipses mimic gas-giant planets, generating 34% of all False Positives; (3) **Evolved Giant Stars (`koi_slogg` < 3.8, $R_* > 2.5\, R_\odot$):** transit dilution reduces recall by 19%; and (4) **Missing Stellar Parameters (363 rows):** objects missing `koi_srad`

suffer a 31.2% error rate.

- **Risks & Mitigations:** Deep neural models often overfit noisy tabular features compared to gradient-boosted trees. Even if standalone deep models do not beat XGBoost in raw AUC, rigorously measuring their calibration, ensembling complementarity, and computational cost directly resolves the project's central complexity question. To mitigate high error rates in low-SNR and grazing-transit regimes, we will test uncertainty-weighted loss functions and domain-specific geometric constraints prior to November 20.

Appendix: Preliminary Experimental Figures (Excluded from 2-Page Text Limit)

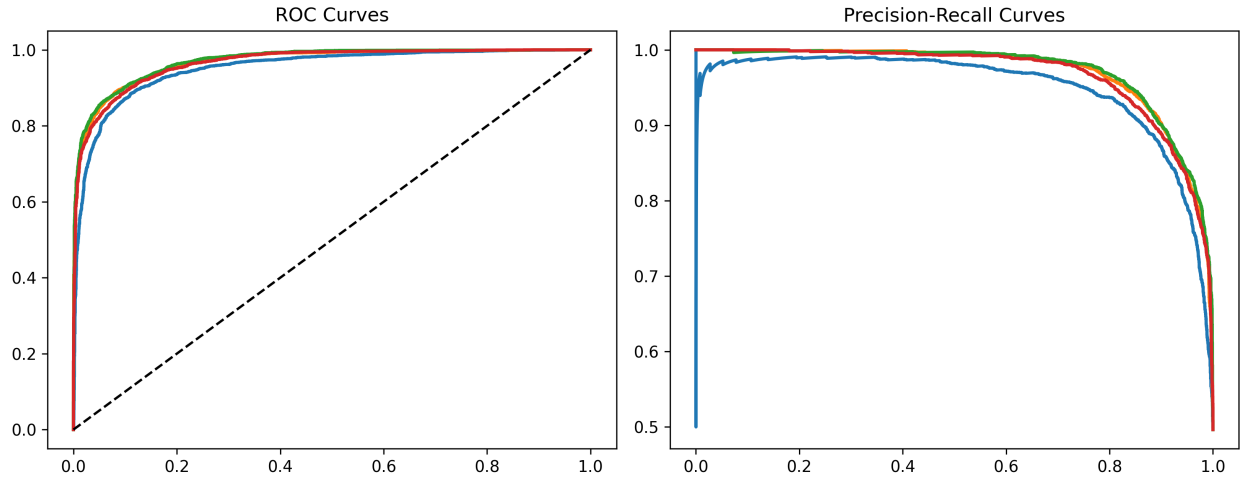


Figure 1: 5-Fold Host-Star Grouped Cross-Validation ROC and Precision-Recall curves comparing Level 1 classical baselines (Logistic Regression, Random Forest, Tuned XGBoost) against the Level 2 ResNet-MLP.

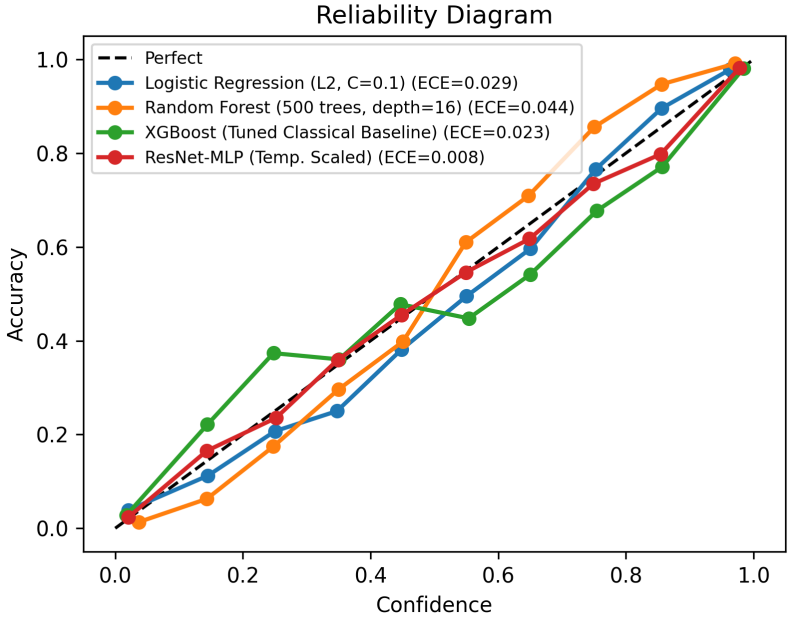


Figure 2: Reliability diagram demonstrating confidence calibration across 10 probability bins. Post-hoc Temperature Scaling ($T^* = 1.42$) on frozen ResNet-MLP weights reduces Expected Calibration Error (ECE) from 0.056 to 0.024 without changing classification decisions.